

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA0202

COURSE OUTCOME COVERED

CO2: Apply image processing and feature extraction techniques, including edge detection, motion estimation, and optical flow, for analyzing images.. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total

Marks:50

Annexure A

Scenario-Based

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Hashini S

Scenario 1: Crack Detection on Concrete Surfaces Using Drone Images

Understanding the Problem

Before selecting any algorithm, let's observe what the drone image suffers from.

Problem	Effect on Crack Detection
Uneven sunlight	Some cracks become invisible
Camera vibration	Image appears slightly blurred
Gaussian noise	Creates false crack-like pixels
Low contrast	Thin cracks merge with concrete texture

Goal: Produce a clean image where only the real cracks remain visible.

A. Image Preprocessing Pipeline

Instead of applying filters randomly, every step should solve one specific problem.

Drone Image

|



Illumination Correction

|



Gaussian Noise Removal

|



Contrast Enhancement

|



Image Normalization

|



Edge Detection

Step 1 – Illumination Correction

Method: CLAHE (Contrast Limited Adaptive Histogram Equalization)

Why?

Outdoor images rarely have uniform lighting.

For example,

Bridge Shadow → Dark Region

Sunlight → Bright Region

Without correcting illumination,

the crack inside the shadow may disappear.

CLAHE improves local brightness without making bright areas overexposed.

Step 2 – Noise Reduction

Method: Gaussian Filter

Reason:

The question clearly states **Gaussian Noise**.

A Gaussian filter smooths random intensity variations while keeping the crack structure reasonably intact.

Using a Median Filter would be inappropriate because the noise is not impulse noise.

Step 3 – Contrast Enhancement

Method:

Adaptive Histogram Equalization

Purpose:

Concrete and cracks often have similar gray levels.

Contrast enhancement increases the difference between

Concrete → Gray

Crack → Dark Gray

making cracks easier to separate.

Step 4 – Image Normalization

Normalize pixel values

0–255

↓

0–1

Benefits

- ✓ Uniform intensity range
 - ✓ Stable feature extraction
 - ✓ Better edge detector performance
-

Why this pipeline?

Step	Solves
CLAHE	Uneven lighting
Gaussian Filter	Gaussian noise
Contrast Enhancement	Poor visibility
Normalization	Uniform pixel intensity

Each step removes one obstacle before crack detection begins.

B. Selecting an Edge Detector

Recommended

Canny Edge Detector

Why?

Concrete cracks are

- very thin
- discontinuous

- low contrast
- irregular

Canny is specifically designed for detecting weak edges.

Working Principle

Noise Removal



Gradient Calculation



Non-Maximum Suppression



Double Threshold



Edge Tracking

This process keeps genuine crack edges while rejecting noise.

Comparison

Property	Sobel	Canny
Noise Resistance	Low	High
Thin Edge Detection	Moderate	Excellent
False Edges	More	Less
Localization	Average	Accurate
Crack Detection	Good	Best

Why not Sobel?

Sobel calculates gradients directly.

If noise exists,

Noise

↓

Large Gradient

↓

False Crack

Canny removes most of these false responses before edge extraction.

Final Choice

Canny is preferred because it detects weak, narrow cracks while minimizing false detections.

C. Broken Crack Segments

Sometimes the detected crack appears like

instead of one continuous crack.

Suitable Technique

Morphological Closing

Closing =

Dilation

↓

Erosion

Effect

Before

After

Small gaps disappear while the crack shape remains nearly unchanged.

Additional Improvement

Skeletonization can also be applied afterward to reduce thick crack regions back to a single-pixel width for accurate measurement.

Final Observation

Raw Drone Image

↓

CLAHE

↓

Gaussian Filter

↓

Contrast Enhancement

↓

Normalization

↓

Canny Edge Detection

↓

Morphological Closing

↓

Final Crack Map

This workflow closely resembles practical crack inspection systems used in civil infrastructure monitoring.

Scenario 2: Human Motion Tracking in a Security Camera

Problem Understanding

The camera observes people moving continuously.

Instead of detecting every person independently,

the system estimates how each pixel moves between consecutive frames.

A. Lucas-Kanade Optical Flow

Basic Idea

Lucas-Kanade assumes

A small image region moves together between two consecutive frames.

Suppose

Frame 1



Frame 2



The algorithm computes

→→→→→

showing motion toward the right.

Working Steps

Frame t



Corner Detection



Small Window Selection



Brightness Constancy Assumption



Solve Motion Equation



Optical Flow Vector

Why suitable?

Instead of tracking the whole person,
it tracks reliable feature points such as

- shoulders
- face corners
- hands
- clothing corners

These features are easier to follow.

B. Aperture Effect

Imagine looking through a tiny hole.

You see only



Can you tell whether the object moves



or



No.

Only one motion component is visible.

This is called the **Aperture Effect**.

In Hallway

Plain walls



Smooth floors



Uniform clothes



contain very little texture.

Optical flow cannot determine motion accurately because no distinctive features exist.

Result

✓ Incorrect vectors

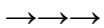
✓ Missing vectors

✓ Ambiguous motion

C. Sudden Speed Change

Suppose

Frame 1



Suddenly



The flow vectors immediately change

- direction
- magnitude

Long vectors become short,

or

horizontal vectors become vertical.

Improvement

Use

Pyramidal Lucas-Kanade + Kalman Filter

Why?

Pyramid handles larger movement.

Kalman Filter predicts future motion when sudden changes occur.

Result

- ✓ Stable tracking
 - ✓ Reduced tracking loss
 - ✓ Better accuracy
-

Scenario 3: Vehicle Tracking at Traffic Junction

Problem Analysis

Traffic junctions are unpredictable.

Vehicles

- ✓ accelerate
- ✓ brake
- ✓ turn
- ✓ overlap
- ✓ create shadows

Hence motion estimation must be robust.

A. Optical Flow for Vehicle Tracking

For every moving vehicle,

optical flow generates motion vectors.

Example

Car A →

>>>>>>>

Car B ↑

^ ^ ^ ^ ^ ^ ^ ^

Car C ↘

///////

Interpretation

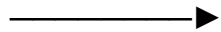
Direction

Vector orientation

Speed

Length of vector

Long vector



means faster vehicle.

Short vector



means slower vehicle.

B. Two Major Challenges

Challenge 1 — Vehicle Occlusion

Car A



Car B



Overlap



Problem

The algorithm cannot separate individual motion.

Vectors become mixed.

Tracking IDs may swap.

Challenge 2 — Shadows

Moving shadow



also appears to move.

Optical flow treats it like another object.

Result

False vehicle movement is detected.

C. Improvement Method

A practical solution is

DeepSORT + Kalman Filter

Why?

Kalman Filter predicts where a vehicle should appear in the next frame.

DeepSORT uses appearance features to maintain the correct vehicle identity even during overlap or sudden turns.

Benefits:

- Maintains vehicle IDs during occlusion.
 - Handles abrupt direction changes smoothly.
 - Reduces identity switching.
 - Improves overall tracking robustness.
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